

Business Model Canvas

1. Need Statement

A **noninvasive, at-home method** is needed to detect **lung cancer** in **high-risk American populations to increase early screening uptake** (currently only 16%) and enable earlier diagnosis when treatments are more effective.

2. Key Partners

- **Technical Advisors:**
- Initial partnerships will involve selecting appropriate **scientific advisors** to aid in paving the swiftest path to market. They will also aid in our reduction of risk and uncertainty when entering the market with novel technology:
 - Xiao-An Fu (University of Louisville): expert in chemical microsensors/microreactors and noninvasive breath analysis for early lung cancer detection
 - Might run into a conflict of interest, since he started and sits on the board of a competitor → maybe find other collaborators of his to aid in development
 - Alan Johnson (UPenn): expert in transport phenomena in nanoscale systems composed of carbon nanotubes, graphene, and other transition metal-based systems that are conjugated with proteins, synthetic peptides, and DNA.
 - Has a patent that measures VOC-based diagnostics for ovarian cancer
 - Not directly associated with any known competitors, might result in a “coopetition” relationship where we may license or adapt some of the tech that Johnson’s group has developed to design the Sundew VOC detector
- **Optimization Partnerships:**
- Optimization partnerships will involve the identification and relationship building with consumer manufacturers and designers to aid in guiding the design iterations to secure a working product, the VOC sensor.
 - Identify and establish relationships with **contract assembly or turnkey contract manufacturing companies** to aid in manufacturing components that will make up the technical aspects of the Sundew device: electrical interface, ideation on the disposable components, and integrating licensed or IP technology related to the aptamer-nanowire (or other modality)
 - This will allow for rapid fabrication and manufacturing of design interactions until an appropriate device modality is identified and cleared.
 - The key components to identify and what assembly companies will be selected will be based on expertise in injection molding for the plastic

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parts for the device, as well as expertise in printed circuit board assembly to support the electrical parts of the device.

- **Professional and Business Relationships:**
- This will involve forging relationships and partnerships with experts in business, finance, and the medical device industries, and understanding how they may be able to support our long-term goals.
 - These will be seasoned business professionals who may eventually be rewarded with a seat at the board if certain metrics are met, such as helping establish manufacturing and engineering relationships, funding and capital acquisition, and aiding in incorporation and the legal aspects of establishing and running a medtech business organization

3. Key Activities

- Sundew would be responsible for the engineering, design, and iterative development of the Sundew VOC detection device. This will involve securing appropriate IP ownership and licensing to enable device success.
- Once a useful and reliable prototype or MVP is identified, Sundew's activities will be involved in leading the industry in non-invasive detection of cancer biomarkers in the form of VOCs. This will allow for industry dominance in the market and will align Sundew as the flagstaff medtech organization in the field
 - This will afford us the competitive advantage in establishing early relationships with the FDA if the device proves successful as an authoritative voice in the field
- Initial activities will involve sensor R&D and prototyping (aptamer selection/enrichment, nanomaterials), sensor fabrication (microfluidics, lab-on-a-chip), device assembly and QC testing, scaling production
- The detection platform and network will depend on the data ecosystem that informs the classification of a given sample that is introduced to the device. This will involve data platform development, device-to-platform communication, clinical partnership building, etc.

4. Key Resources

- **Intellectual**
- Sundew is an early-stage venture and would have to rely heavily on the weight of IP, which would include proprietary aptamer sequences that bind to their respective target molecules and the sensor design, including the sensing processing units.
- Patents and trade secrets would extend to data algorithms and IOP as well as the software and IP for said signal processing and analytics

- **Human**

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- A concerted effort will be made to recruit top engineering talent to be able to create the Sundew device. We will need a dedicated group of biochemical and electrical engineers that aid in aptamer selection in addition to creating an adaptive electrical component that will sustain the support of multiple aptamer cartridges and partial exposure to breath condensate.
- In addition to engineers, we would need a team of software development engineers to streamline the production process of the application that will be conjugated to the device. These folks as well as data scientists and bioinformaticians would be responsible for rolling out a compatible software platform that would determine VOC pattern recognition and establish an AI/ML model for robust VOC classification
- **Financial**
- There would be no business if capital was disregarded. We would have to seek out stable and supportive cash flow to fund and conduct the latter engineering feats. This funding would also help entice high-quality engineers to help in this endeavor.
- It might be advantageous to initially seek non-dilutive funding opportunities. This way, we would not have to gamble with individual investors and we can more assuredly be able to find initial scientific support for the idea if we were to secure non-dilutive funding opportunities in the form of SBIR or STTR

5. Cost Structure

- **Year 1** (\$750k): Aptamer development, breath capture prototype, initial testing
- **Year 2** (\$1M): Semiconductor optimization, usability testing, FDA pre-submission
- **Year 3** (\$750k): Clinical validation, scaled manufacturing prep, market readiness

Category	Fixed Costs	Estimated Fixed Cost	Variable Costs	Estimated Variable Cost
R&D Equipment & Materials	Lab equipment, EG-FET hardware, prototyping tools	\$10,000.00	SELEX kits, disposable VOC sensors, breath cartridges	\$5,000 per SELEX run
Personnel	Core staff salaries for engineering, biology, and product design	\$150,000/year	Contract-based advisors or consultants	\$100/hour
Software & Data Infrastructure	App development, interface programming, initial analytics stack	\$50,000.00	Ongoing maintenance, cloud storage, ML model updates	\$1,000/month
Regulatory/Compliance	FDA 510(k) submission fee	\$12,432.00	Additional validation, audits, and unforeseen testing requirements	\$20,000.00
Manufacturing	Tooling for mold and PCB design	\$20,000.00	Unit cost of consumables like cartridges and aptamer sensors	\$3/unit

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Clinical Testing	Study design, IRB approval, and early trial site partnerships	\$75,000.00	Per-patient testing and sample processing	\$41,000 per patient
Marketing/Go-to-Market Prep	Brand development, investor materials	\$30,000.00	Training programs, pilot site onboarding	\$15,000.00

6. Value Proposition

A **portable, non-invasive, low-cost diagnostic tool** for early lung cancer detection using exhaled breath condensate analyzed via **RNA aptamer biosensors**, offering:

- Earlier diagnosis helps improve outcomes by identifying cancers at an earlier stage
- Broaden early screening access to populations that struggle to travel to screening facilities
- Reduced reliance on expensive, invasive, or radiation-based diagnostics
- Easier, less invasive monitoring of ongoing treatment progress(?)
- Democratize screening beyond high-risk populations to reach the general public(?)

7. Customer Relationships

- **Personalized Support for Healthcare Providers:** Offer dedicated account managers and technical support to hospitals and clinics for seamless integration, training, and ongoing maintenance of the device.
- **Educational Outreach:** Provide workshops, webinars, and detailed documentation to educate healthcare professionals on the benefits of early lung cancer detection using non-invasive technology.
- **Long-Term Partnerships:** Build strategic relationships with key opinion leaders (KOLs) in oncology and pulmonology to foster trust and drive adoption.
- **Community Engagement:** Engage underserved communities through partnerships with local health organizations to promote accessibility and build trust in the technology.
- **Feedback Loops:** Establish regular communication channels (e.g., surveys, focus groups) with early adopters to refine the product based on user experience and clinical outcomes

8. Customer Segments

- **Hospitals and Clinics:** Target large hospitals and community clinics in the U.S. with oncology and pulmonology departments, focusing on point-of-care settings for early lung cancer screening.
- **Diagnostic Laboratories:** Partner with labs seeking innovative, non-invasive diagnostic tools to complement existing lung cancer screening methods.

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- **Underserved Populations:** Reach low-income and rural communities in the U.S. with limited access to advanced diagnostics, addressing healthcare disparities.
- **Oncology Research Institutions:** Engage academic and research institutions for clinical validation studies and to establish credibility in the medical community.
- **Healthcare Payers:** Collaborate with insurance companies and government health programs (e.g., Medicare, Medicaid) to ensure reimbursement for the diagnostic tool, driving adoption.

9. Channels

- **Direct Sales to Healthcare Providers:** Engage hospitals, clinics, and diagnostic labs through direct sales teams to integrate Sundew Dx's breath analysis device into point-of-care settings.
- **Partnerships with Medical Device Distributors:** Collaborate with established medical device distributors to leverage their networks for broader market reach, especially in the U.S. and underserved regions.
- **Online and Digital Platforms:** Utilize a professional website and targeted digital marketing (e.g., LinkedIn, medical conferences, webinars) to educate healthcare professionals and institutions about the technology's benefits.
- **Clinical Partnerships:** Work with oncology-focused research institutions and clinical trial networks to validate and promote the device, building credibility and facilitating adoption.
- **Trade Shows and Medical Conferences:** Showcase the technology at industry events like ASCO (American Society of Clinical Oncology) or RSNA (Radiological Society of North America) to reach key decision-makers.

10. Revenue Streams

- **Device sales:** Diagnostic hardware and disposable cartridges. The goal is to keep the pricing of these tests comparable or slightly less than the \$500 cost of an LDCT Scan/
 - Need for routine screening means patients can buy one device and pay less each interval for cartridges, i.e., Printer/ink model.
- **Institutional partnerships:** Clinical pilot programs and public health screening contracts. Potentially work with high-risk employers to provide early and regular screening to employees or with governments to screen high-risk populations.
- Potential **licensing** of platform technology for other biomarkers/diseases/types of cancer